Proposal: DAMSA at the SNS — a Table-Top Axion Search on a Laser-Stripped Beam-Dump Line

Concept: Install a DAMSA-class detector (sub-meter vacuum decay chamber, two tracking planes, 100-crystal CsI(Tl)+LGAD calorimeter) directly behind a compact tungsten dump on a parasitic, laser-stripped beamline at the SNS linac. Five years of running at **7.3 kW** — 0.3% of SNS beam power, taken without disturbing the neutron program — probes axion-like particles in the open region **$m_a \approx 30$–220 MeV, $g_{a\gamma\gamma} \approx 2\times10^{-6}$–$10^{-2}$ GeV^{-1}**, covering the short-lifetime wedge between the historic beam-dump exclusions and the FASER/PrimEx/BESIII floors. The reach is within a factor 4 in coupling of DAMSA's own 2.5 MW endgame at a facility that does not yet exist — at 0.3% of the beam power, available now.

1. The physics target

Decades of beam dumps (E137, E141, CHARM, NuCal) and collider/Primakoff measurements (PrimEx, GlueX, BESIII, OPAL/LEP, FASER) bracket — but do not close — a wedge of ALP parameter space at m_a 30–300 MeV, $g_{a\gamma\gamma}$ 10^{-5}–10^{-3} GeV^{-1}. It is open for a structural reason: at these couplings the ALP decay length is centimeters to meters, too short to survive the shielding of a conventional dump experiment ("the beam-dump ceiling") and too feeble for colliders. Closing it requires a decay volume centimeters from production — the DAMSA concept: an evacuated chamber where only new physics can produce a displaced two-photon vertex, with invariant mass, dump-pointing, and 50 ps 4D calorimetry replacing passive shielding as the background strategy. The same wedge hosts the QCD-axion-adjacent heavy-ALP bands invoked in dark-sector cosmology; any signal is a discovery with a mass peak and a reconstructible lifetime.

2. Why the SNS is the right host

Parameter	Value	Why it matters
Proton energy	1.3 GeV (post-PPU linac)	0.13 π^0/POT → 0.26 γ/POT; harder spectrum than any sub-GeV driver
Micro-bunch structure	402.5 MHz, 50 ps bunches, 5.9×10^8 p	event-by-event TOF reference
Extraction	laser-assisted H^- stripping, gated: **1 bunch/μs sparse train**	7.3 kW, parasitic; flash from each bunch clears before the next
Exposure	6.4×10^{20} POT/yr → 3.2×10^{21} POT in 5 yr	8×10^{20} photons on the Primakoff target
In-gate duty factor	2×10^{-5} (0.4 ns gate/μs)	steady-state backgrounds negligible

The decisive comparison is **photon flux and timing per dollar**. DAMSA's pathfinder beams are parasitic electron lines rationed in watts (SLAC LESA: 200 W, 1.5×10^{14} EOT baseline); a 1.3 GeV proton delivers 0.26 hard photons for 0.2 nJ, so even 7 kW of SNS beam supplies **35,000× the LESA baseline photon flux**, while the gated stripping laser provides ps-class production timing that no accumulator-based facility can match. No kicker, no ring hardware, no high-power target station: the beamline is an upgrade of the existing SNS laser-stripping development program.

3. Sensitivity, on a controlled footing

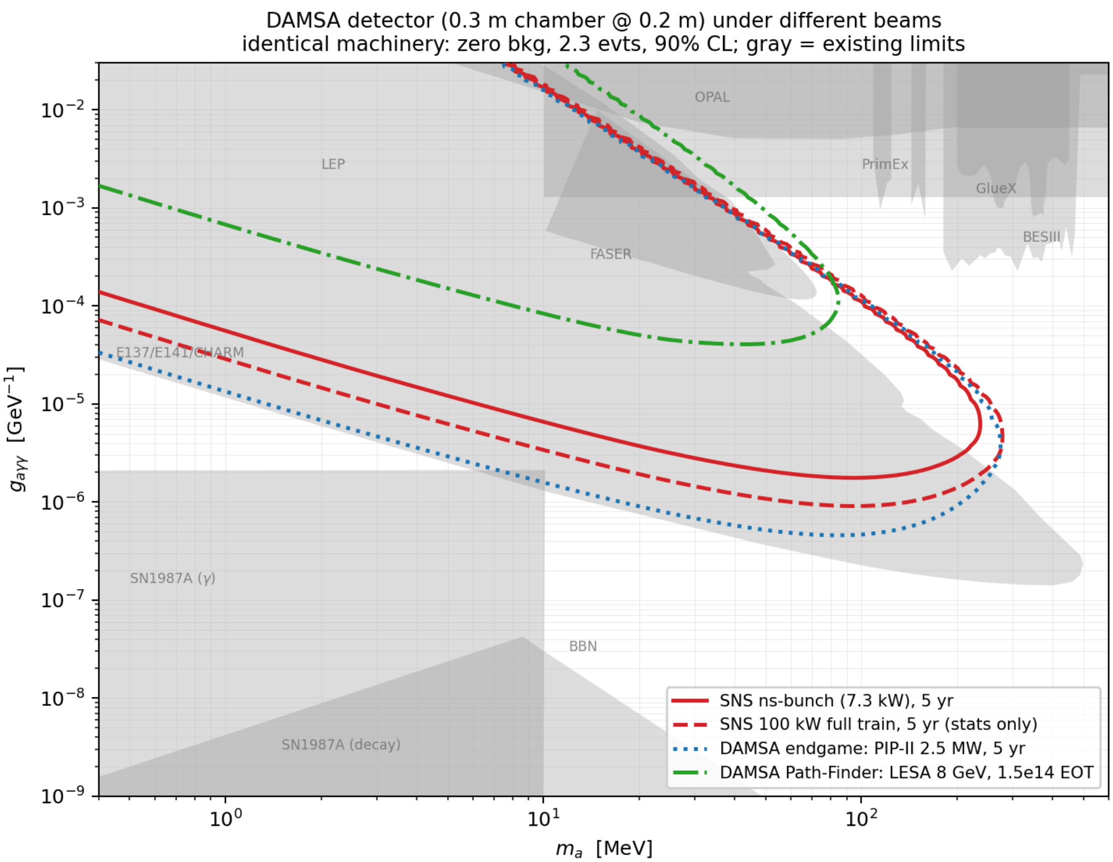

Figure 1: DAMSA detector under four beams

Identical machinery (Primakoff production with Helm form factor + atomic screening; thick-target conversion σ_P/σ_abs; decay-in-flight acceptance; 50% diphoton efficiency; zero background, 2.3 events at 90% CL) applied to the same detector under four beams:

Beam	Exposure	g floor @ 100 MeV	mass reach

SNS sparse ns-bunch, 7.3 kW, 5 yr (this proposal)	3.2×10^{21} POT	**1.8×10^{-6} GeV^{-1}**	221 MeV
SNS 100 kW full train (upgrade path, stats only)	4.5×10^{22} POT	9.4×10^{-7}	258 MeV
DAMSA endgame: PIP-II 2.5 MW (unbuilt)	1.4×10^{24} POT	4.8×10^{-7}	258 MeV
DAMSA Path-Finder: LESA baseline	1.5×10^{14} EOT	no coverage	80 MeV, high-g only

440× less POT than the PIP-II endgame costs only 4× in coupling floor (zero-background limits scale as POT^¼; 1.3 GeV vs 1 GeV adds 1.6× in π^0 yield), while the high-coupling ceiling — set by geometry, not statistics — is essentially identical.

4. Why the projection is trustworthy

The signal model is conservative. Only π^0-decay photons are counted (EM shower and bremsstrahlung photons ignored); π^0 emission is taken isotropic (forward peaking would help a downstream detector); the π^0 yield (0.13/POT) is anchored to the COHERENT/STS flux studies at this exact beam energy; the Primakoff cross section is the standard form with nuclear form factor and atomic screening; efficiency is a flat 50%.

The zero-background assumption is simulated, not asserted. A full Geant4 (11.4, `Shielding` physics list, HP neutron transport) study of this exact geometry — 10 cm W dump, 10 cm (30 X_0) W plug, chamber and calorimeter planes — was used first to reject configurations that fail (a bare dump with the ring's 700 ns pulse is hopeless: 10^{13} in-window particles per pulse), and then to validate this one. The budget per 0.4 ns TOF gate: **zero prompt in-gate hits in 10^5 simulated protons even before the plug** the inter-bunch glow is 11 photons/gate but with a hard nuclear (giant-dipole) endpoint at 25 MeV — below the 30 MeV cluster threshold — plus 0.75 Michel $e^\pm$/gate (charged-vetoed) and 65 invisible neutrinos. Net: ≤26 accidental pairs in 5 yr before topology cuts (an MC-statistics upper limit), **8×10^{-4} expected events after the fiducial-vertex and mass requirements alone**. The three load-bearing design elements are explicit: the 30 MeV per-cluster threshold, the front-tracker charged veto, and the 30 X_0 plug. Every candidate event additionally carries its own ps-level TOF verification against the 402.5 MHz bunch clock.

The beam numbers are SNS parameters, not aspirations: 1.3 GeV/38 mA peak (PPU values), the demonstrated 402.5 MHz micro-bunch structure, and the laser-stripping technique already developed at the SNS. The single accelerator R&D item is gating the stripping laser to select isolated micro-bunches — a laser-timing problem, not a new machine.

5. Cost posture and asks

Detector: table-top (vacuum tank <1 m, two LGAD/µRWELL planes, 100 CsI(Tl) crystals + fast readout) — university-group scale, and identical to hardware the DAMSA collaboration is already prototyping. Beamline: gated stripping laser + transport to a small shielded enclosure with a compact dump. **Asks:** (1) accelerator-physics study of gated laser stripping (sparse micro-bunch extraction at $\geq 10^{-4}$ stripping efficiency); (2) siting study for a 3×3 m enclosure at the linac dump line; (3) high-statistics biased Geant4 run of prompt punch-through (the one budget entry still limited by MC statistics); (4) engagement with the DAMSA collaboration — this is their detector on a beam that advances their endgame physics a decade early.

Supporting material in this repository: beam-structure analysis (*`beam_time_structure_bsm.md`*), RF/timing options (*`rf_structure_options.md`*), sensitivity machinery (*`../study/`*), Geant4 background validation (*`../study/geant/`*), detector comparison (*`damsa_at_sns.md`*).